Multiphysics Eddy-Current Simulation for Efficient Hybrid-Electric Aircraft Propulsion

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The aviation industry is under increasing pressure to enhance fuel efficiency and reduce carbon emissions, driven by the growing demand for global air travel. A promising approach to addressing these challenges is the development of hybrid-electric and electric flight technologies that utilise permanent-magnet synchronous motors to propel aircraft. However, optimising the electromagnetic performance of these systems requires efficient simulations of eddy-current phenomena. A challenging aspect of eddy current formulations is the large null space associated with the curl operator as well as discontinuous material parameters, which further complicate solver performance. We explore work incorporating coupling electromagnetics with thermal effects and fluid-based cooling methods. The results are compared across suitable geometries, emphasising the impact of these coupled effects on solver performance. This analysis again focuses on convergence rates and computational efficiency, aiming to optimise overall system performance.